

CVE VULNERABILITY LOOKUP TOOL – PROJECT SUMMARY

Overview

This project is a Python-based CVE Vulnerability Lookup Tool developed to retrieve and analyze publicly disclosed security vulnerabilities from the National Vulnerability Database (NVD).

The tool queries the NVD API using a CVE (Common Vulnerabilities and Exposures) identifier and displays detailed vulnerability information, including CVSS scores, severity ratings, and vulnerability descriptions. This helps cybersecurity professionals quickly assess security risks and understand the impact of known vulnerabilities.

Features Added

- Automated CVE information retrieval
- Integration with the National Vulnerability Database (NVD) API
- CVSS score extraction and analysis
- Vulnerability severity classification
- Detailed vulnerability description display
- Professional ASCII banner interface
- HTTP request validation and error handling
- User-friendly terminal output
- Real-time vulnerability intelligence gathering

Information Collected

- CVE Identifier
- CVSS Base Score
- Severity Rating
- Vulnerability Description
- NVD Vulnerability Data

Technologies Used

- Python
- Requests Library
- NVD API
- JSON Data Processing

Example Output



```
(root@kali)~[/home/kali/Desktop]
# python cve_lookup.py

  CVE

  CVE VULNERABILITY LOOKUP TOOL

Enter CVE ID (e.g. CVE-2021-44228): CVE-2021-44228

  CVE VULNERABILITY REPORT

CVE ID       : CVE-2021-44228
CVSS Score   : 10.0
Severity     : Critical

Description:
Apache Log4j2 2.0-beta9 through 2.15.0 (excluding security releases 2.12.2, 2.12.3, and 2.3.1) JNDI features used in configuration, log mess
ages, and parameters do not protect against attacker controlled LDAP and other JNDI related endpoints. An attacker who can control log messa
ges or log message parameters can execute arbitrary code loaded from LDAP servers when message lookup substitution is enabled. From log4j 2.
15.0, this behavior has been disabled by default. From version 2.16.0 (along with 2.12.2, 2.12.3, and 2.3.1), this functionality has been co
mpletely removed. Note that this vulnerability is specific to log4j-core and does not affect log4net, log4cxx, or other Apache Logging Servi
ces projects.
```

Concepts Demonstrated

- Vulnerability Management
- CVE Analysis
- Threat Intelligence
- Security Research
- API Integration
- Risk Assessment
- Python Automation
- Cybersecurity Intelligence Gathering

Learning Outcome

This project helps understand how cybersecurity professionals analyze known vulnerabilities and assess their potential impact on systems and applications.

Practical skills gained include:

- CVE Enumeration
- Vulnerability Intelligence Collection
- CVSS Scoring Analysis
- Security Risk Assessment
- API-Based Data Retrieval
- Cybersecurity Automation

Security Applications

- Vulnerability Assessment
- Security Operations Center (SOC) Activities
- Threat Intelligence Research
- Security Monitoring
- Incident Response Preparation
- Risk Management
- Patch Management Prioritization

Future Enhancements

- Search CVEs by Product Name
- Search CVEs by Vendor
- Vulnerability Report Export (PDF/CSV)
- Exploit Availability Detection
- CWE Information Lookup
- EPSS Score Integration
- CVE Trend Analysis
- Threat Intelligence Feed Integration

Disclaimer

This project is intended for educational purposes and authorized security research only. Vulnerability information should be used responsibly to improve security and support defensive cybersecurity activities.

30 Days of Cybersecurity Automation – Day 10

Building practical cybersecurity tools through hands-on automation and real-world security assessments.

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